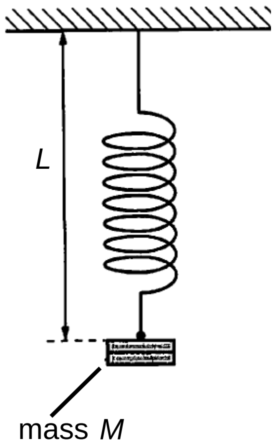
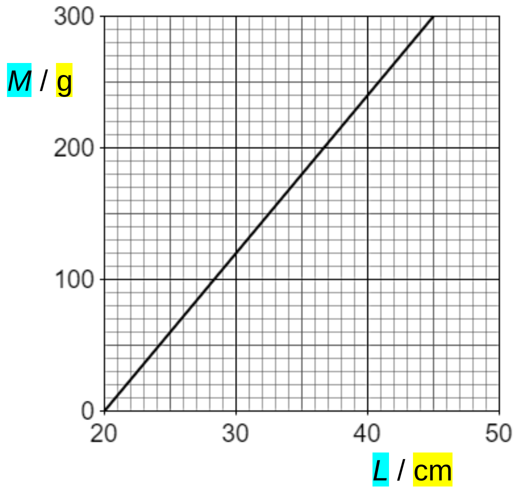
6 One end of a spring is fixed to a support. A mass is attached to the other end of the spring as illustrated below. The variation of mass M with length L is shown in the graph below.





What is the energy stored in the spring when it is extended to a length of 35 cm?

A	0.00750 J	**B**	0.0315 J	**C**	0.132 J	**D**	0.309 J

L2 **Answer: C**

Energy stored in the spring
= Work done by the mass on the spring
= (Area bounded by the graph and horizontal axis) x (acceleration due to gravity)
$= \frac{1}{2}\left(\frac{35-20}{100}\right)\left(\frac{180}{1000}\right)(9.81)$
= 0.132 J

7 A uniform cube of volume 0.729 m^3 is floating in water. The density of water is 1000 kg m^{-3}